

Module3. Detecting Labels in Your Own Video Data

Intelligent Video Analytics with Deep Learning

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Spring, 2019

Module3. Detecting Labels in Your Own Video Data

Learn Python in 5 Minutes

<https://www.stavros.io/tutorials/python/>

Features of Python

- Code is easy to read
- Syntax is concise
- Development speed is fast
- Programming is fun



Package Installation

- Python*

```
pip install  
package_name
```

* Run outside the Python

Using Packages

- Python

```
import  
package_name
```

List

- Python**

```
abc = ['a', 'b',  
      'c']  
abc.append('d')
```

```
print(abc[0])  
print(abc[1])
```

** Also known
as "list"

Dictionary

- Python

```
abc = {}  
abc['a'] = 1  
abc['b'] = 2  
  
print(abc['a']  
])
```


List + Dictionary

- Python

```
li = []  
li.append({'a':1  
})
```

```
print(li[0]  
['a'])
```

String concatenation

- Python

```
abc = ['a', 'b',  
      'c']  
''.join(abc)
```

String substitution

- Python

```
'beat'.replace('b',  
'n')
```

Loop

- Python

```
for i in  
range(1, 5):  
    print(i)
```

Loop

- Python

```
a = range(1,4)
s = [i**2 for i in
a]
print(sum(s))
```

Function

- Python

```
def jiggle(x):  
    return x +  
1
```

File Input

- Python

```
import io

f = io.open('kaist.mp4',
            'rb')
video = f.read()
f.close()
```

Dictionary + List to Table

- Python

```
import pandas as pd

df =
pd.DataFrame([{'a':1, 'b':2},
               {'a':3, 'b':4}])
```

	a	b
0	1	2
1	3	4

Saving a Table as CSV

- Python

```
import pandas as pd

df =
pd.DataFrame([{'a':1, 'b':2},
               {'a':3, 'b':4}])

df.to_csv('test.csv')
```

Google Cloud API

What is an API?

- **API(Application Programming Interface)**
 - API stands for Application Programming Interface, which refers to an interface that allows application programs to control the functions provided by the operating system or programming language.
- **Interface**
 - An interface refers to a shared boundary through which computer systems exchange information.
 - Some computer hardware devices, such as touch screens, can send and receive data through interfaces, while devices like mice and microphones provide interfaces that only transmit data to the system.

REST What is an API?

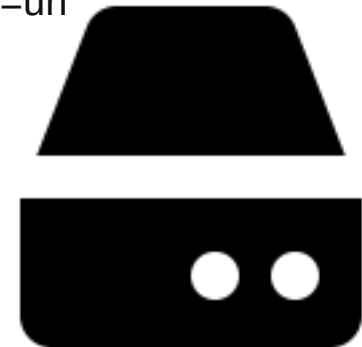
Client - Server
Architecture

1. Request



Resources are represented using
URI format
Ex. <https://www.google.co.kr/search?q=uri>

Actions on resources use HTTP
Methods
Ex. GET, POST



REST What is an API?

Client - Server
Architecture

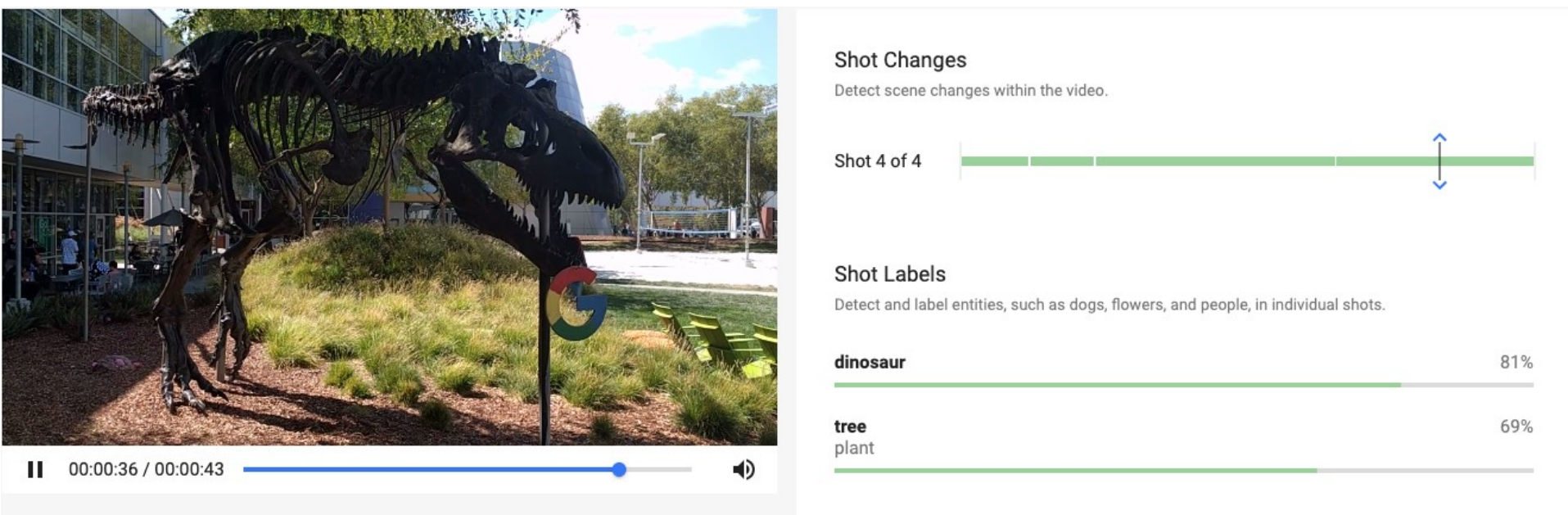
1. Request



2. Response
Result

Google Video Intelligence API

- The API detects the video labels (objects within the video) as well as shot labels (description of the video events over time)



The screenshot displays the Google Video Intelligence API interface. On the left, a video player shows a large dinosaur skeleton sculpture in an outdoor setting. The video progress bar indicates the video is at 00:00:36 of 00:00:43. On the right, the 'Shot Changes' section shows 'Shot 4 of 4' with a green progress bar. Below this, the 'Shot Labels' section lists detected entities: 'dinosaur' with an 81% confidence score and 'tree plant' with a 69% confidence score. Each label has a corresponding green progress bar indicating its duration in the video.

Shot Changes
Detect scene changes within the video.

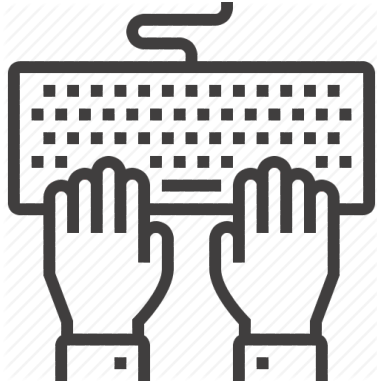
Shot 4 of 4

Shot Labels
Detect and label entities, such as dogs, flowers, and people, in individual shots.

Label	Confidence
dinosaur	81%
tree plant	69%

<https://cloud.google.com/video-intelligence/>

Hands-on-Labs Google Video Intelligence API



Google Video Intelligence API

KAIST_Video_Analytics.ipynb

URL: <https://bit.ly/2IOPeul>

Google Drive Access on Colab

- First Step: Upload your data on Google Drive

<http://drive.google.com>

- Second Step: Enter Authentication Code on Google Colab

Mounting Google Drive locally

The example below shows how to mount your Google Drive in your virtual machine using the code below. Once executed, observe the new file (`foo.txt`) is visible in <https://drive.google.com>

Note this only supports reading and writing files; to programmatically change sharing

```
from google.colab import drive
drive.mount('/content/gdrive')
```

... Go to this URL in a browser: [https://accounts.google.com/o](https://accounts.google.com/o/oauth2/auth)

Enter your authorization code:

- Go to Url
- Type Code

Check Data Available on Google Drive

- Type: `!ls "/content/gdrive/My Drive/"`
 - This is root directory for Google Drive
- For a specific folder, `!ls "/content/gdrive/My Drive/{Your Folder Name}"`

```
# CHECK FILE EXIST  
!ls "{DATA_PATH}"
```

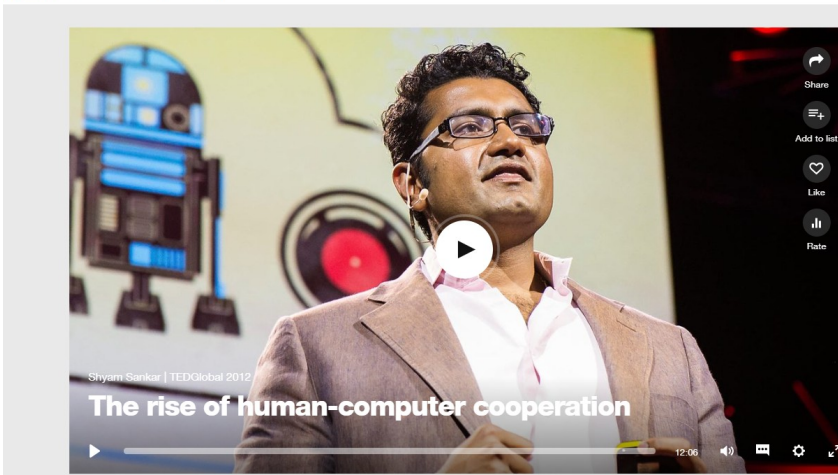
```
google-cloud-sdk-228.0.0-linux-x86_64.tar.gz  key.json  
kaist.mp4                                     segment_df.csv  
KAIST_Video_Analytics.ipynb                  shot_df.csv
```

Conclusion

Let's Cooperate with Deep Learning

- Analytic capability consists of human, computer, and human-computer cooperation.

TED Ideas worth spreading



https://www.ted.com/talks/shyam_sankar_the_rise_of_human_computer_cooperation

“In a freestyle chess tournament in 2005... The surprise came at the end. Who won? Not a grandmaster with a supercomputer, but actually two American amateurs using three relatively weak laptops ...

This is an astonishing result: average men, average machines beating the best man, the best machine. And anyways, isn't it supposed to be man versus machine? **Instead, it's about cooperation, and the right type of cooperation.**”

- Shyam Sankar at TEDGlobal 2012

Thank you □

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References

- Python
 - Eunjung Park, Introduction to Python (on Windows)
 - Taehoon Kim, Python in 140 Minutes